```c

long pthFactor(long n, long p)
{
    int count =0;
    for(long i=1;i<=n;i++)
    {
        if(n%i==0)
        {
            count++;
            if(count==p)
            {
                return i;
            }
        }
    }
    return 0;
}
```

	Test	Expected	Got	
✓	printf("%ld", pthFactor(10, 3))	5	5	✓
✓	printf("%ld", pthFactor(10, 5))	0	0	✓
✓	printf("%ld", pthFactor(1, 1))	1	1	✓

Passed all tests! ✓

Sample Output 2

1

Explanation 2

Factoring n = 1 results in {1}. The p = 1st factor of 1 is returned as the answer.

Answer: (penalty regime: 0 %)

Reset answer

```
/*
 * Complete the 'pthFactor' function below.
 *
 * The function is expected to return a LONG_INTEGER.
 * The function accepts following parameters:
 *  1. LONG_INTEGER n
 *  2. LONG_INTEGER p
 */

long pthFactor(long n, long p)
{
    int count =0;
    for(long i=1;i<=n;i++)
    {
        if(n%i==0)
        {
            count++;
            if(count==p)
            {
                return i;
```

Sample Case 1

Sample Input 1

STDIN   Function

-----   --------

10   →   n = 10

5   →   p = 5

Sample Output 1

0

Explanation 1

Factoring n = 10 results in {1, 2, 5, 10}. There are only 4 factors and p = 5, therefore 0 is returned as the answer.

Sample Case 2

Sample Input 2

STDIN   Function

-----   --------

1   →   n = 1

1   →   p = 1

Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.

The first line contains an integer n, the number to factor.

The second line contains an integer p, the 1-based index of the factor to return.

Sample Case 0

Sample Input 0

```
STDIN     Function
-----     --------
10    →   n = 10
3     →   p = 3
```

Sample Output 0

```
5
```

Explanation 0

Factoring n = 10 results in {1, 2, 5, 10}. Return the p = 3^{rd} factor, 5, as the answer.

sorted ascending. If there is no p^{th} element, return 0.

Example

n = 20

p = 3

The factors of 20 in ascending order are {1, 2, 4, 5, 10, 20}. Using 1-based indexing, if p = 3, then 4 is returned. If p > 6, 0 would be returned.

Function Description

Complete the function pthFactor in the editor below.

pthFactor has the following parameter(s):

int n: the integer whose factors are to be found

int p: the index of the factor to be returned

Returns:

int: the long integer value of the p^{th} integer factor of n or, if there is no factor at that index, then 0 is returned

Constraints

$1 \le n \le 10^{15}$

$1 \le p \le 10^{9}$

```c
 * The function is expected to return an INTEGER.
 * The function accepts INTEGER number as parameter.
 */

int fourthBit(int number)
{
    int binary[32];
    int i=0;
    while(number>0)
    {
        binary[i]=number%2;
        number/=2;
        i++;
    }
    if(i>4)
    {
        return binary[3];
    }
    else
    return 0;
}
```

	Test	Expected	Got	
✓	`printf("%d", fourthBit(32))`	0	0	✓
✓	`printf("%d", fourthBit(77))`	1	1	✓

Passed all tests! ✓

Determine the factors of a number (i.e., all positive integer values that evenly divide into a number) and then return the p^{th} element of the list,

Sample Case 1

Sample Input 1

STDIN Function

----- --------

77 → number = 77

Sample Output 1

1

Explanation 1

- Convert the decimal number 77 to binary number: $77_{10} = (1001101)_2$.
- The value of the 4th index from the right in the binary representation is 1.

Answer: (penalty regime: 0 %)

Reset answer

```
/*
 * Complete the 'fourthBit' function below.
 *
 * The function is expected to return an INTEGER.
 * The function accepts INTEGER number as parameter.
 */
```

Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.

The only line contains an integer, number.

Sample Case 0

Sample Input 0

```
STDIN  Function
-----  --------
32   → number = 32
```

Sample Output 0

```
0
```

Explanation 0

- Convert the decimal number 32 to binary number: $32_{10} = (100000)_2$.
- The value of the 4th index from the right in the binary representation is 0.

A binary number is a combination of 1s and 0s. Its n^{th} least significant digit is the n^{th} digit starting from the right starting with 1. Given a decimal number, convert it to binary and determine the value of the the 4^{th} least significant digit.

Example

number = 23

- Convert the decimal number 23 to binary number: $23^{10} = 2^4 + 2^2 + 2^1 + 2^0 = (10111)_2$.
- The value of the 4^{th} index from the right in the binary representation is 0.

Function Description

Complete the function fourthBit in the editor below.

fourthBit has the following parameter(s):

int number: a decimal integer

Returns:

int: an integer 0 or 1 matching the 4th least significant digit in the binary representation of number.

Constraints

$0 \le number < 2^{31}$

```c
                }
                if(s[i]<'0' || s[i]>'9')
                {
                    flag=0;
                    break;
                }
            }
        }
        else
        flag=0;
        if(flag==1)
        printf("YES\n");
        else
        printf("NO\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✓	3 1234567890 0123456789 0123456.87	YES NO NO	YES NO NO	✓

Passed all tests! ✓

SAMPLE OUTPUT

YES

NO

NO

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<string.h>
int main()
{
    int t;
    scanf("%d",&t);
    while(t--)
    {
        int flag=1;
        char s[100000];
        scanf("%s",s);
        int k=strlen(s);
        if(k==10)
        {
            for(int i=0;i<10;i++)
            {
                if(s[0]=='0')
                {
                    flag=0;
                    break;
                }
                if(s[i]<'0' || s[i]>'9')
                {
                    flag=0;
                    break;
                }
            }
```

, consists of numeric values and it shouldn't have prefix zeroes.

Input:

First line of input is T representing total number of test cases.

Next T line each representing "S" as described in in problem statement.

Output:

Print "YES" if it is valid mobile number else print "NO".

Note: Quotes are for clarity.

Constraints:

1<= T <= 10^3

sum of string length <= 10^5

SAMPLE INPUT

3

1234567890

0123456789

0123456.87

```c
    int max=rate[0];
    char ans[20];
    strcpy(ans,res[0]);
    for(int i=1;i<n;i++)
    {
        if(rate[i]>max)
        {
            max=rate[i];
            strcpy(ans,res[i]);
        }
    }
    printf("%s",ans);
    return 0;
}
```

	Input	Expected	Got	
✓	3 Pizzeria 108 Dominos 145 Pizzapizza 49	Dominos	Dominos	✓

Passed all tests! ✓

These days Bechan Chacha is depressed because his crush gave him list of mobile number some of them are valid and some of them are invalid. Bechan Chacha has special power that he can pick his crush number only if he has valid set of mobile numbers. Help him to determine the valid numbers.

You are given a string "S" and you have to determine whether it is Valid mobile number or not. Mobile number is valid only if it is of length 10

3

Pizzeria 108

Dominos 145

Pizzapizza 49

SAMPLE OUTPUT

Dominos

Explanation

Dominos has maximum points.

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<string.h>
int main()
{
    int n;
    scanf("%d",&n);
    char res[n][21];
    int rate[n];
    for(int i=0;i<n;i++)
    {
        scanf("%s",res[i]);
        scanf("%d",&rate[i]);
    }
    int max=rate[0];
```

by him did not taste good :(. Joey is feeling extremely hungry and wants to eat pizza. But he is confused about the restaurant from where he should order. As always he asks Chandler for help.

Chandler suggests that Joey should give each restaurant some points, and then choose the restaurant having **maximum points**. If more than one restaurant has same points, Joey can choose the one with **lexicographically smallest** name.

Joey has assigned points to all the restaurants, but can't figure out which restaurant satisfies Chandler's criteria. Can you help him out?

Input:

First line has N, the total number of restaurants.

Next N lines contain Name of Restaurant and Points awarded by Joey, separated by a space. Restaurant name has **no spaces**, all lowercase letters and will not be more than 20 characters.

Output:

Print the name of the restaurant that Joey should choose.

Constraints:

1 <= N <= 10^5

1 <= Points <= 10^6

SAMPLE INPUT

```c
            reverse[k]=reverse[size-k-1];
            reverse[size-k-1]=temp;
        }
        for(int j=i+1;j<n;j++)
        {
            if(strcmp(reverse,words[j])==0)
            {
                flag=1;
                break;
            }
        }
        if(flag==1)
        break;
    }
    int len=strlen(reverse);
    printf("%d %c ",len,reverse[len/2]);
    return 0;
}
```

	Input	Expected	Got	
✓	4 abc def feg cba	3 b	3 b	✓

Passed all tests! ✓

Joey loves to eat Pizza. But he is worried as the quality of pizza made by most of the restaurants is deteriorating. The last few pizzas ordered abc

def

feg

cba

SAMPLE OUTPUT

3 b

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<string.h>
int main()
{
    int n,flag=0;
    char temp;
    scanf("%d",&n);
    char words[n][14];
    for(int i=0;i<n;i++)
    scanf("%s",words[i]);
    char reverse[14];
    for(int i=0;i<n-1;i++)
    {
        strcpy(reverse,words[i]);
        int size=strlen(reverse);
        for(int k=0;k<size/2;k++)
        {
            temp=reverse[k];
            reverse[k]=reverse[size-k-1];
```

Danny has a possible list of passwords of Manny's facebook account. All passwords length is odd. But Danny knows that Manny is a big fan of palindromes. So, his password and reverse of his password both should be in the list.

You have to print the length of Manny's password and it's middle character.

Note: The solution will be unique.

INPUT

The first line of input contains the integer N, the number of possible passwords.

Each of the following N lines contains a single word, its length being an odd number greater than 2 and lesser than ***14***. All characters are lowercase letters of the English alphabet.

OUTPUT

The first and only line of output must contain the length of the correct password and its central letter.

CONSTRAINTS

1 ≤ N ≤ 100

SAMPLE INPUT

4

```
        {
            while(str1[i]!=str2[i])
            {
                for(int j=0;j<=i;j++)
                {
                    if(str1[i]<'z')
                    str1[j]++;
                    else
                    {
                        flag=0;
                        break;
                    }
                    if(flag==0)
                    break;
                }
            }
        }
    }
    else
    flag=0;
    if(flag==0)
    printf("NO");
    else
    printf("YES");
    return 0;
}
```

	Input	Expected	Got	
✓	abaca cdbda	YES	YES	✓

Passed all tests! ✓

SAMPLE INPUT

abaca

cdbda

SAMPLE OUTPUT

YES

Explanation

The string ***abaca*** can be converted to ***bcbda*** in one move and to ***cdbda*** in the next move.

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<string.h>
int main()
{
    char str1[1000000],str2[1000000];
    int flag=1;
    scanf("%s",str1);
    scanf("%s",str2);
    int a= strlen(str1);
    int b = strlen(str2);
    if(a==b)
    {
        for(int i=a-1;i>=0;i--)
        {
```

Two strings ***A*** and ***B*** comprising of lower case English letters are compatible if they are equal or can be made equal by following this step any number of times:

· Select a prefix from the string ***A*** (possibly empty), and increase the alphabetical value of all the characters in the prefix by the same valid amount. For example, if the string is ***xyz*** and we select the prefix ***xy*** then we can convert it to ***yx*** by increasing the alphabetical value by 1. But if we select the prefix ***xyz*** then we cannot increase the alphabetical value.

Your task is to determine if given strings ***A*** and ***B*** are compatible.

Input format

First line: String ***A***

Next line: String ***B***

Output format

For each test case, print ***YES*** if string ***A*** can be converted to string ***B***, otherwise print ***NO***.

Constraints

1 ≤ len(A) ≤ 1000000

1 ≤ len(B) ≤ 1000000

```
    scanf("%s",str2);
    while(str1[i]!='\0')
    {
        count1++;
        i++;
    }
    while(str2[j]!='\0')
    {
        count2++;
        j++;
    }
    printf("%d %d\n",count1,count2);
    printf("%s%s\n",str1,str2);
    t=str1[0];
    str1[0]=str2[0];
    str2[0]=t;
    printf("%s %s",str1,str2);
    return 0;
}
```

	Input	Expected	Got	
✓	abcd ef	4 2 abcdef ebcd af	4 2 abcdef ebcd af	✓

Passed all tests! ✓

4 2

abcdef

ebcd af

Explanation

a = "abcd"

b = "ef"

|a| = 4

|b| = 2

a + b = "abcdef"

a' = "ebcd"

b' = "af"

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main()
{
    char str1[10],str2[10],t;
    int i=0,j=0;
    int count1=0,count2=0;
    scanf("%s",str1);
    scanf("%s",str2);
    while(str1[i]!='\0')
    {
        count1++;
        i++;
```

		is fun	is fun	

Passed all tests! ✓

Input Format

You are given two strings, ***a*** and ***b***, separated by a new line. Each string will consist of lower case Latin characters ('a'-'z').

Output Format

In the first line print two space-separated integers, representing the length of ***a*** and ***b*** respectively.

In the second line print the string produced by concatenating ***a*** and ***b*** (***a*** + ***b***).

In the third line print two strings separated by a space, ***a'*** and ***b'***. ***a'*** and ***b'*** are the same as ***a*** and ***b***, respectively, except that their first characters are swapped.

Sample Input

abcd

ef

Sample Output

Explanation 0

In the given string, there are three words ["This", "is", "C"]. We have to print each of these words in a new line.

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main()
{
    char s[1000];
    scanf("%[^\n]s",s);
    for(int i=0;s[i]!='\0';i++)
    {
        if(s[i]!=' ')
        printf("%c",s[i]);
        else
        printf("\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✓	This is C	This is C	This is C	✓

Given a sentence, ***s***, print each word of the sentence in a new line.

Input Format

The first and only line contains a sentence, ***s***.

Constraints

$1 \le len(s) \le 1000$

Output Format

Print each word of the sentence in a new line.

Sample Input 0

This is C

Sample Output 0

This

is

C

```c
    while(t--)
    {
        char str[100000];
        int count=0;
        scanf("%s",str);
        for(int i=0;str[i]!='\0';i++)
        {
            char c= str[i];
            if((c=='a') || (c=='e') || (c=='i') || (c=='o') || (c=='u') || (c=='A') || (c=='E') || (c=='I') || (c==
            count++;
        }
        printf("%d\n",count);
    }
    return 0;
}
```

	Input	Expected	Got	
✓	2 nBBZLaosnm JHkIsnZtTL	2 1	2 1	✓
✓	2 nBBZLaosnm JHkIsnZtTL	2 1	2 1	✓

Passed all tests! ✓

Constraints:

1 ≤ T ≤ 10
1 ≤ length of string ≤ 10^5

SAMPLE INPUT

2

nBBZLaosnm

JHkIsnZtTL

SAMPLE OUTPUT

2

1

Explanation

In test case 1, a and o are the only vowels. So, count=2

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main()
{
    int t;
    scanf("%d",&t);
```

	Input	Expected	Got	
✓	a11472o5t6	0 2 1 0 1 1 1 1 0 0	0 2 1 0 1 1 1 1 0 0	✓
✓	lw4n88j12n1	0 2 1 0 1 0 0 0 2 0	0 2 1 0 1 0 0 0 2 0	✓
✓	1v88886l256338ar0ekk	1 1 1 2 0 1 2 0 5 0	1 1 1 2 0 1 2 0 5 0	✓

Passed all tests! ✓

Today, Monk went for a walk in a garden. There are many trees in the garden and each tree has an English alphabet on it. While Monk was walking, he noticed that all trees with vowels on it are not in good state. He decided to take care of them. So, he asked you to tell him the count of such trees in the garden.

Note: The following letters are vowels: 'A', 'E', 'I', 'O', 'U', 'a', 'e', 'i', 'o' and 'u'.

Input:

The first line consists of an integer *T* denoting the number of test cases.

Each test case consists of only one string, each character of string denoting the alphabet (may be lowercase or uppercase) on a tree in the garden.

Output:

For each test case, print the count in a new line.

Explanation 0

In the given string:

- ***1*** occurs two times.
- ***2, 4, 5, 6*** and ***7*** occur one time each.

The remaining digits ***0, 3, 8*** and ***9*** don't occur at all.

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main()
{
    char str[1000];
    scanf("%s",str);
    int hash[10]={0,0,0,0,0,0,0,0,0,0,};
    int temp;
    for(int i=0;str[i]!='\0';i++)
    {
        temp=str[i]-'0';
        if(temp<=9 && temp>=0)
        {
            hash[temp]++;
        }
    }
    for(int i=0;i<=9;i++)
    {
        printf("%d ",hash[i]);
    }
    return 0;
}
```

Given a string, ***s***, consisting of alphabets and digits, find the frequency of each digit in the given string.

Input Format

The first line contains a string, ***num*** which is the given number.

Constraints

1 ≤ len(num) ≤ 1000

All the elements of num are made of English alphabets and digits.

Output Format

Print ten space-separated integers in a single line denoting the frequency of each digit from ***0*** to ***9***.

Sample Input 0

a11472o5t6

Sample Output 0

0 2 1 0 1 1 1 1 0 0

```
        for(j=1;j<1001;j++)
        {
            if(arr[i][j]<0)
            total+=arr[i][j];
        }
    }
    printf("%lld\n",(-1)*total);
    return 0;
}
```

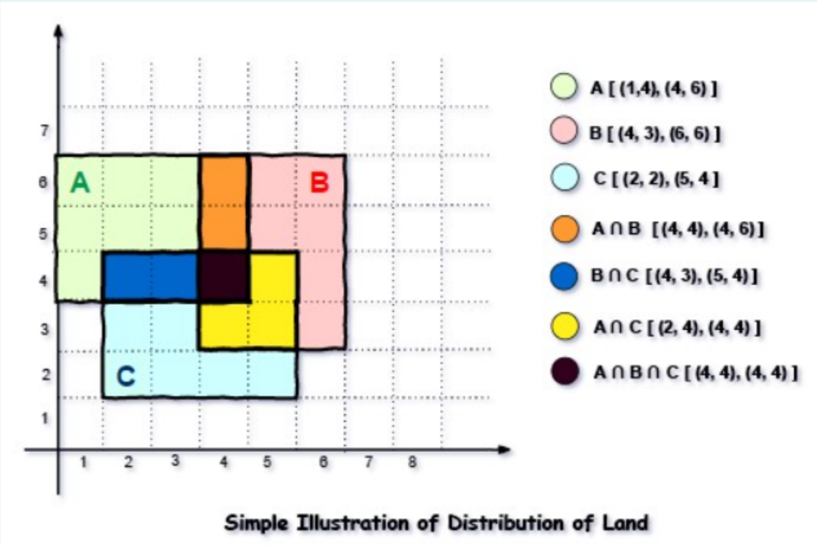
	Input	Expected	Got	
✓	3 1 4 4 6 1 4 3 6 6 2 2 2 5 4 3	35	35	✓
✓	1 48 12 49 27 8	0	0	✓
✓	3 88 34 99 76 44 82 65 94 100 81 58 16 65 66 7	10500	10500	✓

Passed all tests! ✓

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main()
{
    int i,j,n,x1,x2,y1,y2,t=0;
    long long total=0;
    int arr[1001][1001]={0};
    scanf("%d",&n);
    while(n--)
    {
        scanf("%d %d %d %d %d",&x1,&y1,&x2,&y2,&t);
        for(i=x1;i<=x2;i++)
        {
            for(j=y1;j<=y2;j++)
            {
                if(arr[i][j]==0)
                arr[i][j]+=t;
                else if(arr[i][j]>0)
                arr[i][j]=(-1)*(arr[i][j]+t);
                else if(arr[i][j]<0)
                arr[i][j]-=t;
            }
        }
    }
    for(i=1;i<1001;i++)
    {
        for(j=1;j<1001;j++)
        {
            if(arr[i][j]<0)
            total+=arr[i][j];
        }
    }
    printf("%lld\n",(-1)*total);
    return 0;
}
```

Explanation



For given sample input (see given graph for reference), compensation money for different farmers is as follows:

Farmer with land area A: $C_1 = 5 * 1 = 5$

Farmer with land area B: $C_2 = 6 * 2 = 12$

Farmer with land area C: $C_3 = 6 * 3 = 18$

Total Compensation Money = $C_1 + C_2 + C_3 = 5 + 12 + 18 = 35$

(X1, Y1) and ***(X2, Y2)*** are the locations of first and last square block on the diagonal of the rectangular region.

Output Format:

Print the total amount he has to return to farmers to solve the conflict.

Constraints:

1 ≤ N ≤ 100

1 ≤ X1 ≤ X2 ≤ 1000

1 ≤ Y1 ≤ Y2 ≤ 1000

1 ≤ C ≤ 1000

SAMPLE INPUT

3

1 4 4 6 1

4 3 6 6 2

2 2 5 4 3

SAMPLE OUTPUT

35

	1 12 0 12 1 12 1 12 0 12 1 12			

Passed all tests! ✓

Shyam Lal, a wealthy landlord from the state of Rajasthan, being an old fellow and tired of doing hard work, decided to sell all his farmland and to live rest of his life with that money. No other farmer is rich enough to buy all his land so he decided to partition the land into rectangular plots of different sizes with different cost per unit area. So, he sold these plots to the farmers but made a mistake. Being illiterate, he made partitions that could be overlapping. When the farmers came to know about it, they ran to him for compensation of extra money they paid to him. So, he decided to return all the money to the farmers of that land which was overlapping with other farmer's land to settle down the conflict. All the portion of conflicted land will be taken back by the landlord.

To decide the total compensation, he has to calculate the total amount of money to return back to farmers with the same cost they had purchased from him. Suppose, Shyam Lal has a total land area of ***1000 x 1000*** equal square blocks where each block is equivalent to a unit square area which can be represented on the co-ordinate axis. Now find the total amount of money, he has to return to the farmers. Help Shyam Lal to accomplish this task.

Input Format:

The first line of the input contains an integer ***N***, denoting the total number of land pieces he had distributed. Next ***N*** line contains the ***5*** space separated integers ***(X1, Y1)***, ***(X2, Y2)*** to represent a rectangular piece of land, and cost per unit area ***C***.

	Input	Expected	Got	
✓	5 0 3 1 6 0 2 0 7 1 15	7 3 2 15 6	7 3 2 15 6	✓
✓	6 0 1 0 26 0 39 0 37 0 7 0 13	39 37 26 13 7 1	39 37 26 13 7 1	✓
✓	12 1 12 1 14 1 18 1 1 1 2 1 3 1 5 1 8 1 9 1 10 0 29 0 31	31 29 18 14 12 10 9 8 5 3 2 1	31 29 18 14 12 10 9 8 5 3 2 1	✓
✓	12 0 12 1 12 0 12 1 12	12 12 12 12 12 12 12 12 12 12 12 12	12 12 12 12 12 12 12 12 12 12 12 12	✓

```c
#include<stdio.h>
struct data
{
    int gen;int tal;
};
int main()
{
    int n;
    scanf("%d",&n);
    struct data a[n];
    for(int i=0;i<n;i++)
    scanf("%d %d",&a[i].gen,&a[i].tal);
    for(int i=0;i<n-1;i++)
    {
        for(int j=0;j<n-i-1;++j)
        {
            if(a[j].tal<a[j+1].tal)
            {
                struct data temp=a[j];
                a[j]=a[j+1];
                a[j+1]=temp;
            }
        }
    }
    for(int i=0;i<n;i++)
    {
        if(a[i].gen==0)
        printf("%d ",a[i].tal);
    }
    for(int i=0;i<n;i++)
    {
        if(a[i].gen==1)
        printf("%d ",a[i].tal);
    }
}
```

1 <= N <= 10^5

0 <= ai <= 1

1 <= bi <= 10^9

Output Format

Output space-separated integers, which first contains the talent levels of all female candidates sorted in descending order and then the talent levels of male candidates in descending order.

SAMPLE INPUT

5

0 3

1 6

0 2

0 7

1 15

SAMPLE OUTPUT

7 3 2 15 6

Answer: (penalty regime: 0 %)

✓	1 2 3 4 5 6 7 8 9	25 20	25 20	✓
✓	21 422 423 443 586 645 657 846 904	2591 2356	2591 2356	✓

Passed all tests! ✓

Microsoft has come to hire interns from your college. N students got shortlisted out of which few were males and a few females. All the students have been assigned talent levels. Smaller the talent level, lesser is your chance to be selected. Microsoft wants to create the result list where it wants the candidates sorted according to their talent levels, but there is a catch. This time Microsoft wants to hire female candidates first and then male candidates.

The task is to create a list where first all-female candidates are sorted in a descending order and then male candidates are sorted in a descending order.

Input Format

The first line contains an integer N denoting the number of students. Next, N lines contain two space-separated integers, ai and bi.

The first integer, ai will be either 1(for a male candidate) or 0(for female candidate).

The second integer, bi will be the candidate's talent level.

Constraints

25

20

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main()
{
    int arr[3][3];
    for(int i=0;i<3;i++)
    {
    for(int j=0;j<3;j++)
    {
        scanf("%d",&arr[i][j]);
    }
    }
    int odd=0,even=0;
    for(int i=0;i<3;i++)
    {
        for(int j=0;j<3;j++)
        {
            if((i+j)%2!=0)
            odd+=arr[i][j];
            else
            even+=arr[i][j];
        }
    }
    printf("%d\n%d",even,odd);
}
```

	Input	Expected	Got	

You are given a two-dimensional 3*3 array starting from A [0][0]. You should add the alternate elements of the array a Time left 1:29:54 Hide
should print two different numbers the first being sum of A 0 0, A 0 2, A 1 1, A 2 0, A 2 2 and A 0 1, A 1 0, A 1 2, A 2 1.

Input Format

First and only line contains the value of array separated by single space.

A 0 0	**A 0 1**	**A 0 2**
4	6	9
A 1 0	**A 1 1**	**A 1 2**
2	5	8
A 2 0	**A 2 1**	**A 2 2**
1	3	7

Output Format

First line should print sum of A 0 0, A 0 2, A 1 1, A 2 0, A 2 2

Second line should print sum of A 0 1, A 1 0, A 1 2, A 2 1

SAMPLE INPUT

1 2 3 4 5 6 7 8 9